Standard Flight Procedure

APPN Aerial SOP — Locked revision 1.0

Standard Flight Procedure

Version 0.1 – DRAFT, April 2026

> **Important**
>
> This protocol outlines the standard operational flight procedure for routine APPN aerial surveys using the GOBI or CALViS payloads. It covers the three most common setup scenarios, which can be adapted to suit most field situations. The equipment list under each scenario covers field equipment only.
>
> For sensor-specific equipment, procedures, and safety information, see the relevant sensor fieldbook:
>
> - CALViS Fieldbook
> - GOBI M350 Fieldbook
> - GOBI IF1200 Fieldbook
>
> For periodic calibration / APEx test flights, see the Validation Flight Procedure.

> **Important**
>
> **Document status — work in progress.** This protocol is a draft and requires further revision before it can be considered final. New figures also need to be produced to illustrate the procedures described below.
>
> **Outstanding TODOs:**
>
> - [ ] Review and revise all section content for technical accuracy and completeness.
> - [ ] Produce new figures illustrating:
> - [ ] Flight-line layout and overlap geometry
> - [ ] Flight-line layout for the two panel setup
> - [ ] Reflectance panel placement and orientation
> - [ ] GCP distribution across the survey area
> - [ ] On-site equipment setup
> - [ ] Cross-check terminology and links against the sensor fieldbooks.
> - [ ] Confirm equipment checklist with field operators.

Background and common ellments of all standard flights

> **Important**
>
> This section describes the **placement of GCPs and reflectance panels** that is common to all standard flight patterns described below.
>
> All placements given here (distances, positions, panel locations, GCP distributions) are **approximate guidelines, not strict requirements**. Operators should use common sense and adapt the layout to the realities of the site:

- Use the **nearest existing path, bare strips, tramlines etc** in the crop to access and place the panels and GCP's.
- **Do not place panels or GCPs inside experimental plots** or anywhere they would damage the crop, disturb a treatment, or obstruct other field operations.
- Keep panels and GCPs clear of shadows (from trees, infrastructure, the operator, or the UAV itself) and away from the take-off / landing zone.
- Where the ideal position is not accessible, move to the closest practical location that still satisfies the intent of the layout (e.g. good distribution across the survey area, panels overflown at correct time).

Caution

Keep panels and GCPs away from the edge of the capture window.

Depending on the exact flight parameters (altitude, overlap, turn geometry), GOBI and CALViS can fail to capture data the outermost portion of the planned capture polygon — meaning targets placed near the edge may be missed entirely or only partially captured.

- **Place all GCPs and reflectance panels inside an "effective capture area" approximately 10% smaller than the planned capture polygon** (i.e. set back from each edge by ~10% of the polygon's short-axis dimension).
- If site access forces a panel or GCP to sit near the edge of the field, **expand the planned capture polygon outward** so that the target still falls comfortably inside the effective capture area.
- Do not rely on the nominal polygon boundary as the usable imaging extent.

Failure to place ELM panels correctly can render a flight's data completely unusable for radiometric analysis.

Flight-line Orentiation

Note

GRYFN recommends aligning flight lines with the planting direction for "operational reasons", as this greatly reduces stitching artefacts between adjacent flight lines. Testing flight-line angle is a top priority for APEx, and this guidance may be revised before the start of the season. TODO: ADJUST THIS BEFORE SEASON STARTS

Panel Setup

All reflectance panels must be deployed on **open, unshaded, level ground**, clear of the take-off / landing zone and clear of any obstructions that could cast shadows across the panels at any point during the flight window.

TODO: Insert image of a panel set correctly laid out on a folding table.

Place panels on a folding table. All radiometric panels deployed in the field should be placed on a folding table (e.g. [TODO: add link to recommended Bunnings table]). Mounting the panels on a table is important because:

- **Levelling.** Panels must be level for accurate reflectance retrieval, and it is far easier to level a table than to find a level patch of ploughed or uneven ground.
- **Reduced spectral contamination.** Elevating the panels above the canopy reduces stray reflected light from the surrounding crop contaminating the panel signal.
- **Reduced dust and debris.** Raising the panels off the ground reduces their susceptibility to dust, soil splash, and trampling damage.

Level the table using the spirit level, and **record the height of the panel surface above the ground** in the flight log.

Panel ordering and orientation. Lay out each panel set in **order of brightness** (e.g. darkest → brightest), with the row of panels **oriented along the planned flight direction**.

> **Important**
>
> Hyperspectral line scanners image perpendicular to the flight direction. If panels are laid out perpendicular to flight (i.e. across the scan line), multiple panels can fall within a single scanline and introduce significant errors during the ELM calibration. Always align the panel row **with** the flight direction so each panel occupies its own set of scanlines.

Validation Panel Setup In all the flight designs described below, the **2-panel validation panel set** is deployed near the **middle of the survey area**, on its own levelled folding table, following the same general setup rules as the main ELM panels above.

In addition, **two GCPs are paired with the validation panels** to support data QC:

1. **Elevated GCP** — placed on the validation panel table, alongside the validation panels.
2. **Ground GCP** — placed directly on the ground in the **adjacent flight line**, positioned **perpendicular to the elevated GCP** (i.e. on a line running across-track from the table).

   TODO: Insert image showing the validation panel table with the elevated GCP, and the paired ground GCP in the adjacent flight line.

This paired GCP arrangement is used during data QC to check for:

- **Flight-line tearing / mis-registration** between adjacent lines (the two GCPs should resolve to the same horizontal position across-track).
- **Vertical (height) accuracy**, by comparing the measured elevation difference between the elevated and ground GCPs against the recorded table height.

> **Caution**
>
> This procedure was designed with **Propeller Aeropoints** in mind. If using a different GCP system, consider whether it is safe and practical to take a location measurement on top of a table. If not, place the "elevated" GCP on the ground a few metres clear of the table instead, and record the offset.

Ground Control Points

All of the standard flight designs specify a **minimum of 5 GCPs**. This number is based on the standard pack of 5 Propeller Aeropoints.

There are several distinct types of spatial error in UAV imagery, and the placement of these five GCPs is chosen so that **at least one GCP checks each of the main error types**:

- **2 × centre GCPs (the validation pair, described above)** — check for **flight-line tearing** (across-track mis-registration) and **vertical / elevation accuracy**.
- **2 × corner GCPs** — placed near **opposite corners** of the survey area, **directly under a flight line**, to check for **temporal GNSS drift** across the duration of the flight.
- **1 × off-axis GCP** — placed **between two flight lines** (i.e. not directly under the nadir track) to check for **off-axis / lateral registration errors**.

> **Note**
>
> The 5-GCP layout is a **minimum**. Additional GCPs are encouraged for all flights and are **strongly recommended** for larger survey areas and for the multi-flight capture configuration (see below).

Dual ELM panel flight

Overview

This flight uses **two sets of ELM (Empirical Line Method) reflectance panels**, deployed at approximately the **1/3 and 2/3 positions along the long axis of the survey area**.

Rationale:

- **Reduced temporal offset between target and panel capture.** Splitting the panels across the field minimises the elapsed time (and therefore the change in solar geometry and atmospheric conditions) between any given image and its nearest panel overflight.
- **Redundancy.** Both panel sets can be ingested by the GRYFN Processing Tool (GPT) during radiometric calibration, providing a backup if one panel set is shaded, disturbed, or otherwise unusable. *(TODO: cross-link to the processing pipeline page once published.)*
- **Improved reflectance retrieval** over longer flights where illumination conditions are more likely to drift.

When to use this flight:

- **Default** flight pattern for all nodes equipped with two ELM panel sets.
- **Recommended** for all flights longer than **10 minutes**.
- **Mandatory** for flights longer than **15 minutes**.
- **Mandatory** under variable illumination conditions (e.g. broken cloud), regardless of flight duration. (APEx results dependant)

Flight design

> TODO: Insert annotated diagram showing:
>
> - Panel placement at the 1/3 and 2/3 positions along the long axis.
> - Flight-line orientation relative to the panels.
> - Recommended buffer between panels and the Edge of the capture pologyon.
> - GCP placment

Ground Equipment Required

- ☐ 2 × GRYFN 4-panel reflectance set (11%, 30%, 56%, 82%)
- ☐ 1 × GRYFN 2-panel validation reflectance set
- ☐ 5 × Ground control points (GCPs) — Propeller Aeropoints or equivalent (more for larger areas)
- ☐ 2 × Folding tables (1 per panel set)
- ☐ Spirit level (to level the tables)
- ☐ Tape measure

> **Important**
>
> All other equipment listed in the relevant sensor fieldbook is also required.

Single ELM panel flight

Overview

This flight uses a **single set of ELM (Empirical Line Method) reflectance panels**, deployed near the **center of the survey area** and are only flown over once.

Rationale:

- **Simpler logistics.** Only one main panel set needs to be deployed, levelled, and monitored — appropriate for nodes equipped with a single ELM panel set or for short missions where setup time is at a premium.
- **Adequate radiometric calibration for short flights.** Over short durations, illumination conditions are unlikely to drift far enough to require a second panel observation, and the validation panels provide a basic check on the calibration.

When to use this flight:

- **Default** flight pattern for nodes equipped with **only one ELM panel set**.

- **Acceptable** for short missions of **10 minutes** under stable illumination conditions (clear sky or uniform overcast).
- **Not recommended** for flights longer than 10 minutes — use the Dual ELM panel flight instead.
- **Not permitted** under variable illumination (e.g. broken cloud), regardless of flight duration. (TODO: If APEx allows this)

Flight design

TODO: Insert annotated diagram showing:

- Panel placement near center of the survey area.
- Validation panel placement in the middle of the survey area.
- Flight-line orientation relative to the panels.
- Recommended buffer between panels and the edge of the capture polygon.
- GCP placement (5-GCP layout described above).

Ground Equipment Required

☐ 1 × GRYFN 4-panel reflectance set (11%, 30%, 56%, 82%)
☐ 1 × GRYFN 2-panel validation reflectance set
☐ 5 × Ground control points (GCPs) — Propeller Aeropoints or equivalent (more for larger areas)
☐ 2 × Folding tables (1 per panel set)
☐ Spirit level (to level the tables)
☐ Tape measure

> **Important**
>
> All other equipment listed in the relevant sensor fieldbook is also required.

Multi-Flight Capture

Overview

This configuration covers survey areas that are **too large to be captured in a single mission** (due to battery endurance, radio range, or regulatory line-of-sight constraints) and must be split into **two or more sub-flights**. Sub-flights are deliberately planned to **overlap** along their shared boundary, and a **shared ELM panel set plus a pair of "tie" GCPs are placed inside the overlap region** and **left in place between sub-flights**. This shared equipment is overflown by both sub-flights and is the primary mechanism for stitching the sub-flights into a single, radiometrically and geometrically consistent product.

Rationale:

- **Shared radiometric reference across sub-flights.** Because the same ELM panel set is observed by both sub-flights (without being moved or re-levelled), it provides a common radiometric anchor, letting the sub-flights be calibrated onto a consistent reflectance scale even when illumination changes between flights.
- **Geometric tie between sub-flights.** The two shared "tie" GCPs in the overlap zone are observed by both sub-flights and pin the sub-flights together horizontally and vertically, preventing seams and offsets at the sub-flight boundary.
- **Equipment stays put.** Leaving the shared panels and tie GCPs in place between sub-flights eliminates re-deployment error (panels re-levelled differently, GCPs moved by a few cm) that would otherwise corrupt the tie.
  Battery / payload swaps and any required pilot relocations happen while the shared equipment remains undisturbed.
  (Operators should remain on-site to monitor the shared equipment between sub-flights.)\

When to use this flight:

- **Mandatory** when the planned survey area cannot be completed in a single mission within UAV endurance and regulatory line-of-sight limits.

- **Recommended** when a single-mission survey would exceed **15 minutes** on the wing and the area can be sensibly divided into shorter sub-flights.
- Each individual sub-flight should otherwise follow the Dual ELM panel flight layout, with its own panels at ~1/3 and 2/3 along its long axis (one of which will be the shared overlap panel set).

Flight design

TODO: Insert annotated diagram showing:

- Two (or more) sub-flight polygons with a deliberate overlap zone at their shared boundary.
- The **shared ELM panel set and 2 × tie GCPs** located inside the overlap zone, overflown by both sub-flights.
- Dual ELM panel placement within each sub-flight (the shared set serves as one of the two panel positions for each sub-flight).
- Validation panel placement within one of the sub-flights, per the Validation Panel Setup above.
- Expanded GCP distribution across the full survey area.

Ground Equipment Required

- ☐ 2 × GRYFN 4-panel reflectance set (11%, 30%, 56%, 82%) — 1 dedicated to the individual sub-flights + **1 shared set placed in the overlap zone and left in place between sub-flights**
- ☐ 1 × GRYFN 2-panel validation reflectance set
- ☐ 10 × Ground control points (GCPs) — Propeller Aeropoints or equivalent (minimum; more strongly recommended for large areas). Includes **2 × "tie" GCPs placed inside the overlap zone and left in place between sub-flights**
- ☐ 4 × Folding tables (1 per panel set)
- ☐ Spirit level (to level the tables)
- ☐ Tape measure

> **Important**
>
> All other equipment listed in the relevant sensor fieldbook is also required.

> **Caution**
>
> The shared overlap-zone panel set and tie GCPs **must not be moved, re-levelled, or otherwise disturbed between sub-flights**. If they are bumped, shaded, or relocated, the radiometric and geometric tie between sub-flights is lost and the sub-flights cannot be reliably stitched. If the shared equipment is disturbed, treat the affected sub-flights as independent captures and re-fly if possible.

> **Note**
>
> **Two-sub-flight limit for nodes with only 2 ELM panel sets.**
>
> The configuration described above assumes a node has **2 ELM panel sets** in total: one is dedicated to the first sub-flight and the other is the shared set in the overlap zone (which then also serves as one of the panel positions for the second sub-flight). This covers a survey split into **exactly 2 sub-flights**.
>
> Splitting a survey into **3 or more sub-flights** is not possible with only 2 panel sets, because every internal (middle) sub-flight has **two overlap zones** (one with each neighbour) and therefore requires its own pair of shared panel sets.
>
> If a node with only 2 panel sets must cover an area larger than two sub-flights can fit, **divide the area into independent zones, each sized to fit within a 2-sub-flight capture**, and treat each zone as a separate site. Do **not** attempt a 3+ sub-flight capture by re-using or re-deploying a panel set during sub-flights — moving the shared equipment breaks the radiometric and geometric tie that the multi-flight design depends on.
